

Task 9: Calibration Statistics, Correlation Analysis and Feature Engineering

Objective

The objective of this project was to analyze calibration measurement data by computing replicate statistics, performing correlation analysis, fitting calibration models, generating visualizations, and engineering new features for machine learning applications.

Dataset

The dataset contains calibration measurements collected from three engineering domains:

- Biochemistry
- Electronics
- Mechanical Engineering

Each record contains:

- Domain
- Input Value
- Signal
- Temperature
- Stress
- Replicate measurements

Methodology

The project was completed in four stages:

1. Replicate Statistics

Calculated:

- Mean
- Standard Deviation
- Coefficient of Variation
- 95% Confidence Interval

Each experiment was classified as:

- Stable
- Moderate
- Unstable

2. Correlation Analysis

Performed:

- Pearson Correlation
- Spearman Correlation

between calibration variables.

3. Calibration Analysis

A linear calibration model was fitted using NumPy.

The following evaluation metrics were computed:

- R^2 Score
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

Calibration curves were generated for every engineering domain.

4. Feature Engineering

Additional machine-learning features were created:

- Input Squared
- Input Cubed
- Signal/Input Ratio
- Input–Temperature Interaction

Categorical variables were encoded to generate a machine-learning-ready dataset.

Output Files

Generated CSV files:

- replicate_summary.csv
- correlation_summary.csv
- calibration_summary.csv
- engineered_features.csv
- ml_ready_dataset.csv

Generated figures:

- correlation_signal_input.png
- biochem_calibration_curve.png
- electronics_calibration_curve.png
- mechanical_calibration_curve.png

Libraries Used

- pandas

- numpy
- scipy
- matplotlib

Conclusion

This project successfully implemented a complete calibration data analysis workflow. Statistical analysis, correlation analysis, calibration modeling, visualization, and feature engineering were performed successfully, resulting in structured datasets and graphical outputs suitable for further machine learning applications.